II. EDA-Data Import and Export

AIM:

- Importing data from CSV, Excel, SQL databases, and web scraping
- Handling different data formats
- Export a DataFrame to an Excel file.

PROCEDURE:

1. Import pandas, sqlite3, requests and BeautifulSoup.
2. Import data from CSV using pd.read_csv().
3. Import data from Excel using pd.read_excel().
4. Create/connect to SQLite database, create table, insert records and retrieve using pd.read_sql_query().
5. Extract HTML table using requests, BeautifulSoup and pd.read_html().
6. Handle different data formats using pd.to_datetime(), astype(), etc.
7. Export the DataFrame using to_excel().

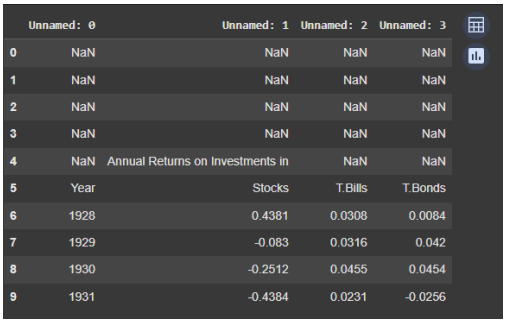
PROGRAM AND OUTPUT:

```
import pandas as pd
df_csv = pd.read_csv("/content/test_Y3wMUE5_7gLdaTN.csv")
df_csv.head()
```

	Sno.	Candidate	Party	Criminal Cases	Education	Age	Total Assets	Liabilities
0	1	A Nepolian	IND	0	Post Graduate	40	Rs 67,355	Rs 0
1	2	A. Babu Mailan	Naam Tamilar Katchi	0	Others	54	Rs 36,62,000	Rs 46,58,722
2	3	A. Jayakumar	IND	0	Graduate	43	Rs 6,02,959	Rs 0
3	5	A. Nizar Ahamed	IND	0	Graduate	57	Rs 19,84,500	Rs 0
4	6	A. Rajaram	Republican Party of India (Athawale)	0	Graduate Professional	51	Rs 5,500	Rs 0

```
df_excel = pd.read_excel("/content/Historicalinvesttemp.xlsx",
sheet_name="Sheet1")
```

df_excel.head()



	Unnamed: 0	Unnamed: 1	Unnamed: 2	Unnamed: 3
0	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN
2	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN
4	NaN	Annual Returns on Investments in	NaN	NaN
5	Year	Stocks	T.Bills	T.Bonds
6	1928	0.4381	0.0308	0.0084
7	1929	-0.083	0.0316	0.042
8	1930	-0.2512	0.0455	0.0454
9	1931	-0.4384	0.0231	-0.0256

```
import sqlite3
import pandas as pd
conn = sqlite3.connect('my_database.db')
cursor = conn.cursor()

cursor.execute('''
CREATE TABLE IF NOT EXISTS students (
    id INTEGER PRIMARY KEY,
    name TEXT,
    major TEXT,
    gpa REAL,
    enrollment_date TEXT
)
''')

cursor.executemany('''
INSERT INTO students (id, name, major, gpa, enrollment_date) VALUES (?, ?,
?, ?, ?)
''', [
    (1, 'Nina',      'Computer Science',        3.8, '2021-08-15'),
    (2, "Lerin",   'Mechanical Engineering',  3.4, '2020-08-20'),
    (3, 'Sofia',   'AIML',                3.9, '2022-01-10'),
    (4, 'Krithick',        'AIDS',              3.2, '2019-08-25'),
    (5, 'Bhagat',  'Computer Science',         3.6, '2021-01-12'),
])

conn.commit()
print("Database and 'students' table created with sample data.")
sql_df = pd.read_sql_query("SELECT * FROM students", conn)
print(sql_df)
```

```
Database and 'students' table created with sample data.
   id      name                  major  gpa enrollment_date
0   1      Nina       Computer Science  3.8      2021-08-15
1   2     Lerin  Mechanical Engineering  3.4      2020-08-20
2   3     Sofia                   AIML  3.9      2022-01-10
3   4  Krithick                   AIDS  3.2      2019-08-25
4   5    Bhagat       Computer Science  3.6      2021-01-12
```

df=pd.read_excel("/content/Historicalinvesttemp.xlsx")

df.to_excel("/content/weeeeee.xlsx", index=False)

df.head()

	Unnamed: 0	Unnamed: 1	Unnamed: 2	Unnamed: 3
0	NaN	NaN	NaN	NaN
1	NaN	NaN	NaN	NaN
2	NaN	NaN	NaN	NaN
3	NaN	NaN	NaN	NaN
4	NaN	Annual Returns on Investments in	NaN	NaN

RESULT:

Thus, the given programs are written and executed successfully.